Global Wholesale Distributor

Technical Documentation

1. Project Overview

The **DE Mini Project** is a multi-layer data engineering pipeline built using the **Medallion** Architecture (Bronze → Silver → Gold). It processes raw business data into clean, structured, and analytics-ready datasets to generate meaningful business insights and KPIs.

2. Architecture: 3-Layer Medallion Design

Bronze Layer (azure_bronze)

Purpose:
Raw data ingestion from Azure sources.

Processing:

- Removal of Fivetran metadata columns:
  - _file
  - _line
  - _modified
  - _fivetran_synced

Tables (7):

- customers
- exchange_rates
- invoice_line_items
- invoices
- payments
- products
- regions

Notebook:

- nb_bronze

Silver Layer (transformed_tables)

Purpose:
Data cleaning and transformation.

Processing:

- Deduplication
- Null handling:
    - Strings → "unknown"
    - Numeric values → 0
- Date parsing (to_date standardization)
- Fact table creation using joins and business logic

Tables (8):

- Same as Bronze tables
- fact_sales (enriched with revenue and profit calculations)

Notebooks:

- nb_silver_common
- nb_customers
- nb_invoices
- nb_invoice_line_items
- nb_products

Gold Layer (analytical_tables)

Purpose:
Analytics-ready dimensional model.

Tables (6):

Dimensions:

- dim_customers
- dim_products
- dim_regions

Facts:

- fact_sales
- fact_payments

Cube:

- cube_sales_analytics (aggregated metrics)

Notebooks:

- nb_fact_dim
- nb_cube_table
- nb_kpi

3. Key Business Metrics (KPIs)

Calculated in nb_kpi:

- **Total Revenue:** $9.75M USD
- **Total Quantity Sold:** 27,081 units
- **Average Invoice Value:** $6,555.24 USD
- **Top Customers:**

    - Corporate segment
    - Retail segment
- **Top Products:**
    - Electronics
    - Accessories
- **Cancellation Rate:**
    - 25.6% (512 / 2,000 invoices)
- **Discount Impact:**
    - $382K USD total discounts
    - 7.49% average discount
- **Regional Performance:**
    - WEST region's highest revenue ($2.85M)
- **Customer Transaction Patterns:**
    - Invoice frequency analysis

4. Data Flow

Azure Sources →	Bronze (Raw) → Silver (Cleaned) →	Gold (Analytics)
↓	↓	↓
Metadata Removal	Deduplication Null Handling Date Parsing Calculations	Dimensions Facts KPIs

5. Technical Stack

- **Platform:** Databricks Unity Catalog
- **Language:** PySpark + SQL
- **Storage Format:** Delta Lake (with column mapping enabled)

- **Transformations:** DataFrame API + SQL (DDL/DML)

6. Summary

This project demonstrates end-to-end data engineering best practices, including:

- Clear separation of layers (Bronze, Silver, Gold)
- Data quality handling (nulls, deduplication, standardization)
- Scalable transformations using PySpark and SQL
- Business-focused analytics and KPI generation